

Q9. Name Anticoagulant Drugs, Uses, and Precautions in Dental Practice

1. Warfarin

- **Use:**
 - Prevents **thrombosis** in conditions like **atrial fibrillation**, **deep vein thrombosis (DVT)**, and **pulmonary embolism (PE)**.
 - Used in patients with **mechanical heart valves**.
- **Precautions in Dental Practice:**
 - **Monitor INR** (International Normalized Ratio) to ensure **safe clotting levels**.
 - Avoid **invasive dental procedures** if INR is high (greater than 3).
 - Ensure **good oral hygiene** to avoid **gingival bleeding**.
 - Avoid **prolonged bleeding** post-extraction by using **pressure packs**.
 - Consider **local hemostatic measures** (e.g., **gel foam**, **hemostatic sponges**) after surgery.

2. Heparin

- **Use:**
 - Used in hospital settings for **acute thrombosis**, **myocardial infarction**, and **post-surgical** patients.
 - **Low molecular weight heparin** (e.g., **enoxaparin**) is often used in outpatient settings.
- **Precautions in Dental Practice:**
 - If patients are on **heparin therapy**, consult with their **physician** about stopping it prior to dental procedures.
 - **Monitor for bleeding complications**, especially in invasive procedures like extractions.
 - **Avoid trauma** to mucous membranes to reduce the risk of excessive bleeding.

3. Direct Oral Anticoagulants (DOACs)

- **Examples: Apixaban, Rivaroxaban, Dabigatran**
- **Use:**
 - Used for prevention and treatment of **DVT**, **PE**, and **atrial fibrillation**.
 - Often preferred due to **no routine monitoring** required (e.g., **INR** is not needed).
- **Precautions in Dental Practice:**
 - **Timing of last dose:** Dental procedures should be scheduled at least **12-24 hours** after the last dose to minimize bleeding risk.
 - **Monitor for prolonged bleeding** post-extraction and use **local hemostatic agents** as needed.

- Consult with the physician if the patient is on **Dabigatran** and requires a procedure, as it can have a longer half-life.

4. Aspirin

- **Use:**
 - Used as a **blood thinner** in patients at risk for **heart attacks** and **strokes**.
 - **Low-dose aspirin** is often used for **cardiovascular disease prevention**.
- **Precautions in Dental Practice:**
 - **Aspirin** is a **platelet inhibitor** and may increase bleeding during dental procedures, especially after **extractions**.
 - **Discontinue** aspirin **7 days before** elective dental surgery** to reduce bleeding risk.
 - For **high-risk cardiac patients**, the benefit of continuing aspirin often outweighs the risk of bleeding, so **consult with the physician**.

5. Clopidogrel (Plavix)

- **Use:**
 - Used in patients who have had a **stent placement, myocardial infarction, or stroke** to prevent **thrombosis**.
- **Precautions in Dental Practice:**
 - **Avoid invasive procedures** immediately after starting Clopidogrel, as it **inhibits platelet aggregation**.
 - **Consult with the physician** about stopping Clopidogrel prior to dental work, especially for procedures like **oral surgery** or **extractions**.
 - **Monitor for bleeding**, especially in the case of **tooth extractions** and **post-operative care**.

6. Dipyridamole

- **Use:**
 - Often combined with **aspirin** for **stroke prevention** and in patients with **prosthetic heart valves**.
- **Precautions in Dental Practice:**
 - **Bleeding risk** is increased with **dipyridamole**, especially in invasive procedures.
 - **Monitor bleeding times** and apply appropriate **hemostatic measures** post-surgery.

General Precautions in Dental Practice for Anticoagulants

- **Thorough medical history:** Always ask the patient about anticoagulant therapy during **pre-treatment evaluation**.
- **Consult with physician:** In cases of **high-risk patients** on anticoagulants, always **consult with their physician** before performing invasive procedures.
- **Local hemostatic measures:** Use measures like **pressure, sutures**, and **hemostatic agents** (e.g., **gelatin sponges, bone wax**) to control bleeding.

Post-procedure care: Advise patients to **avoid excessive physical activity** post-surgery to minimize the risk of bleeding

Q13. Syncope (Fainting)

- **Definition:** Generalized weakness of muscles, loss of postural tone, inability to stand, and loss of consciousness.
- **Faintness:** A prodromal phase of syncope, with symptoms like light-headedness, blurring of vision, heaviness in the lower limbs, and eventual loss of consciousness.

Mechanism of Syncope

- **Fainting** is due to **cerebral ischemia**, especially affecting the **brainstem**.
- **Faintness** occurs due to insufficient ischemia to impair consciousness. Causes include sudden **vasodilatation**, a **sudden fall in cardiac output**, or both.

Types of Syncope

1. Vasovagal Syncope:

- **Mechanism:** Reflex slowing of the heart through the **vagus nerve**, with a fall in **arterial pressure** and **peripheral vascular resistance**.
- **Causes:** Emotional stress, warm overcrowded rooms, sudden pain, mild blood loss, anemia, fever, and fasting.
- **Clinical features:**
 - Prodromal phase: **nausea, sweating, yawning, epigastric distress, tachypnoea**, weakness, and confusion.
 - Followed by: **faintness, pallor, coldness of hands and feet**, and loss of consciousness.
- **Examination:**
 - Tachycardia in the prodromal phase, then **bradycardia, low blood pressure**, and **weak pulse**.
 - **Consciousness is regained rapidly**.
- **Common in young people**.

2. Cardiac Syncope:

- **Cause:** Sudden reduction in cardiac output, can be due to the heart beating too fast, too slow, or not at all.

Common causes:

- **Complete heart block** (Stokes-Adams-Morgagni syndrome)
- **Paroxysmal tachycardias** (supraventricular or ventricular)
- **Sick sinus syndrome**
- **Acute massive myocardial infarction**
- **Aortic stenosis** (exertional syncope)
- **Hypertrophic cardiomyopathy** (exertional syncope)
- **Primary pulmonary hypertension**
- **Pulmonary embolism**
- **Left atrial ball valve thrombus** or **myxoma**
- **Tetralogy of Fallot**

3. Orthostatic Syncope (Postural Syncope):

- **Cause:** Occurs when standing up quickly or standing still for long, leading to **postural hypotension** due to loss of vasoconstrictor reflexes in the lower limb vessels.
- **Common causes:**
 - **Physiological.**
 - **Idiopathic orthostatic hypotension.**
 - **Drugs:** Antihypertensive drugs, **ACE inhibitors, beta-blockers, calcium channel blockers, diuretics, nitrates.**
 - **Diabetic neuropathy** and **tabes dorsalis.**
 - **Haemorrhage.**
 - **Vasodilators** and **volume depletion.**

4. Syncope Associated with Cerebrovascular Disease:

- **Cause: Occlusion** in large neck arteries reduces blood flow to the brainstem, especially during physical activity, causing loss of consciousness.

5. Miscellaneous Causes of Syncope:

- **Carotid Sinus Syncope:** Compression of the **carotid sinus** (e.g., turning the head, tight collar, shaving) in elderly patients.
- **Vagal and glossopharyngeal neuralgia.**
- **Micturition syncope:** Occurs during or after micturition in elderly, especially after rising from bed.
- **Cough syncope:** Seen in elderly with chronic bronchitis after paroxysms of cough.
- **Hypoglycaemia.**

Hyperventilation and hysterical fainting